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June 19, 2024

Docket No.: 50-348  
50-364

NL-24-0235

U. S. Nuclear Regulatory Commission  
ATTN: Document Control Desk  
Washington, D. C. 20555-0001

Joseph M. Farley Nuclear Plant - Unit 1 & 2  
Licensee Event Report 2024-002-01  
Manual Reactor Trip due to Loss of Power to the 2A 125 Volt DC Distribution Panel

Ladies and Gentlemen:

Following the completion of a substantial hazard evaluation Southern Nuclear Company is submitting the enclosed revised Licensee Event Report for Unit 1 & 2 in accordance with the requirements of 10 CFR 21.2(c) and 10 CFR 50.73(a)(2)(iv)(A).

This letter contains no NRC commitments. If you have any questions regarding this submittal, please contact Gene Surber, Licensing Manager, at (334) 661-2265.

Respectfully submitted,

A handwritten signature in dark ink, appearing to read "Edwin Dean III", with a stylized flourish at the end.

Edwin Dean III

Vice President – Farley

ED/rgs/cbg

Enclosure: Unit 1 & 2 Licensee Event Report 2024-002-01

Cc: Regional Administrator, Region II  
NRR Project Manager – Farley Nuclear Plant  
Senior Resident Inspector – Farley Nuclear Plant  
RTYPE: CFA04.054

**Joseph M. Farley Nuclear Plant - Unit 1 & 2  
Licensee Event Report 2024-002-01  
Manual Reactor Trip due to Loss of Power to the 2A 125 Volt DC Distribution Panel**

**Enclosure**

**Unit 1 & 2 Licensee Event Report 2024-002-01**



## LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form)

<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to [Infocollections.Resource@nrc.gov](mailto:Infocollections.Resource@nrc.gov), and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

## 1. Facility Name

Joseph M. Farley Nuclear Plant, Unit 2

☒ 050  
☐ 052

## 2. Docket Number

364

## 3. Page

1 OF 3

## 4. Title

Manual Reactor Trip due to Loss of Power to the 2A 125 Volt DC Distribution Panel

## 5. Event Date

## 6. LER Number

## 7. Report Date

## 8. Other Facilities Involved

| Month | Day | Year | Year | Sequential Number | Revision No. | Month | Day | Year | Facility Name | <input checked="" type="checkbox"/> 050 | Docket Number |
|-------|-----|------|------|-------------------|--------------|-------|-----|------|---------------|-----------------------------------------|---------------|
| 02    | 16  | 2024 | 2024 | - 002 -           | 01           | 06    | 19  | 2024 | FNP Unit 1    |                                         | 348           |
|       |     |      |      |                   |              |       |     |      | Facility Name | <input type="checkbox"/> 052            | Docket Number |

## 9. Operating Mode

1

## 10. Power Level

100

## 11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

|                                                                          |                                                    |                                                    |                                                        |                                                    |                                     |
|--------------------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------|--------------------------------------------------------|----------------------------------------------------|-------------------------------------|
| <input checked="" type="checkbox"/> 10 CFR Part 20                       | <input type="checkbox"/> 20.2203(a)(2)(vi)         | <input checked="" type="checkbox"/> 10 CFR Part 50 | <input type="checkbox"/> 50.73(a)(2)(ii)(A)            | <input type="checkbox"/> 50.73(a)(2)(viii)(A)      | <input type="checkbox"/> 73.1200(a) |
| <input type="checkbox"/> 20.2201(b)                                      | <input type="checkbox"/> 20.2203(a)(3)(i)          | <input type="checkbox"/> 50.36(c)(1)(i)(A)         | <input type="checkbox"/> 50.73(a)(2)(ii)(B)            | <input type="checkbox"/> 50.73(a)(2)(viii)(B)      | <input type="checkbox"/> 73.1200(b) |
| <input type="checkbox"/> 20.2201(d)                                      | <input type="checkbox"/> 20.2203(a)(3)(ii)         | <input type="checkbox"/> 50.36(c)(1)(ii)(A)        | <input type="checkbox"/> 50.73(a)(2)(iii)              | <input type="checkbox"/> 50.73(a)(2)(ix)(A)        | <input type="checkbox"/> 73.1200(c) |
| <input type="checkbox"/> 20.2203(a)(1)                                   | <input type="checkbox"/> 20.2203(a)(4)             | <input type="checkbox"/> 50.36(c)(2)               | <input checked="" type="checkbox"/> 50.73(a)(2)(iv)(A) | <input type="checkbox"/> 50.73(a)(2)(x)            | <input type="checkbox"/> 73.1200(d) |
| <input type="checkbox"/> 20.2203(a)(2)(i)                                | <input checked="" type="checkbox"/> 10 CFR Part 21 | <input type="checkbox"/> 50.46(a)(3)(ii)           | <input type="checkbox"/> 50.73(a)(2)(v)(A)             | <input checked="" type="checkbox"/> 10 CFR Part 73 | <input type="checkbox"/> 73.1200(e) |
| <input type="checkbox"/> 20.2203(a)(2)(ii)                               | <input checked="" type="checkbox"/> 21.2(c)        | <input type="checkbox"/> 50.69(g)                  | <input type="checkbox"/> 50.73(a)(2)(v)(B)             | <input type="checkbox"/> 73.77(a)(1)               | <input type="checkbox"/> 73.1200(f) |
| <input type="checkbox"/> 20.2203(a)(2)(iii)                              |                                                    | <input type="checkbox"/> 50.73(a)(2)(i)(A)         | <input type="checkbox"/> 50.73(a)(2)(v)(C)             | <input type="checkbox"/> 73.77(a)(2)(i)            | <input type="checkbox"/> 73.1200(g) |
| <input type="checkbox"/> 20.2203(a)(2)(iv)                               |                                                    | <input type="checkbox"/> 50.73(a)(2)(i)(B)         | <input type="checkbox"/> 50.73(a)(2)(v)(D)             | <input type="checkbox"/> 73.77(a)(2)(ii)           | <input type="checkbox"/> 73.1200(h) |
| <input type="checkbox"/> 20.2203(a)(2)(v)                                |                                                    | <input type="checkbox"/> 50.73(a)(2)(i)(C)         | <input type="checkbox"/> 50.73(a)(2)(vii)              |                                                    |                                     |
| <input type="checkbox"/> OTHER (Specify here, in abstract, or NRC 366A). |                                                    |                                                    |                                                        |                                                    |                                     |

## 12. Licensee Contact for this LER

## Licensee Contact

Gene Surber, Farley Licensing Manager

## Phone Number (Include area code)

3346612265

## 13. Complete One Line for each Component Failure Described in this Report

| Cause | System | Component | Manufacturer | Reportable to IRIS | Cause | System | Component | Manufacturer | Reportable to IRIS |
|-------|--------|-----------|--------------|--------------------|-------|--------|-----------|--------------|--------------------|
| B     | EJ     | BKR       | S188         | Y                  |       |        |           |              |                    |

## 14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

## 15. Expected Submission Date

| Month | Day | Year |
|-------|-----|------|
|       |     |      |

## 16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On February 16, 2024, at 00:48 CDT while in Mode 1 and at 100% power, Farley Nuclear Plant (FNP) operators manually tripped Unit 2 following a loss of control of critical air-operated valves due to a partial loss of A-train DC power. LA08-2, the feeder breaker to the 2A 125VDC Distribution Panel, was found tripped open. This event resulted in a loss of letdown and the reactor makeup system. With volume control tank (VCT) makeup capability lost, operators made the decision to trip the reactor prior to charging pump suction rollover to the refueling water storage tank (RWST).

The cause of the breaker opening was due to a short between wires on the A phase current transformer (CT) / sensor wiring harness in breaker LA08-2 because of poor workmanship. Immediate corrective action included replacing breaker LA08-2 with a spare breaker to restore plant equipment to operation and stabilize the plant.

This event is reportable under 10 CFR 50.73(a)(2)(iv)(A) due to the automatic actuation of multiple systems listed in 10 CFR 50.73(a)(2)(iv)(B). FNP Unit 1 was not affected during this event. However, the 10 CFR 21 notification does apply to Unit 1.

**LICENSEE EVENT REPORT (LER)  
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form  
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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|                                        |                                                                                       |                         |                      |                          |                |
|----------------------------------------|---------------------------------------------------------------------------------------|-------------------------|----------------------|--------------------------|----------------|
| <b>1. FACILITY NAME</b>                | <input checked="" type="checkbox"/> <b>050</b><br><input type="checkbox"/> <b>052</b> | <b>2. DOCKET NUMBER</b> | <b>3. LER NUMBER</b> |                          |                |
| Joseph M. Farley Nuclear Plant, Unit 2 |                                                                                       | 364                     | <b>YEAR</b>          | <b>SEQUENTIAL NUMBER</b> | <b>REV NO.</b> |
|                                        |                                                                                       |                         | 2024                 | 002                      | 01             |

**NARRATIVE****EVENT DESCRIPTION:**

On February 16, 2024, at 00:48 CDT, with reactor power at 100% and in Mode 1, Farley Nuclear Plant (FNP) operators manually tripped Unit 2 following a partial loss of A-train 125V DC power which affected critical air-operated valves. Breaker LA08-2 [EIS: EJ/BKR] the supply breaker to the 2A 125VDC Distribution Panel, was found tripped open. Loss of power to the 2A 125VDC Distribution Panel resulted in a loss of letdown and the reactor makeup system. With the volume control tank (VCT) [EIS: CB / TK] makeup capability lost operators made the decision to trip the reactor prior to the charging pump suction rollover to the refueling water storage tank (RWST).

Operating events and actions occurred in the following timeline on February 16, 2024:

00:35 Multiple Main Control Room Alarms are received.  
00:35 Breaker LA08-02 Tripped (Entry into TS 3.8.9)  
00:35 Entry into Abnormal Operating Procedure for CVCS Malfunction  
00:36 Entry into Abnormal Operating Procedure of Loss of Instrument Air  
00:36 Letdown confirmed isolated and loss of makeup capability.  
00:48 Operators Manually Trip Reactor based on VCT level.  
01:42 Reactor Coolant Pump 2B secured due to low seal leak off.  
01:45 Pressurizer (PZR) level rises to Technical Specification (TS) high value of 63.5%. (Entry into TS 3.4.9)  
02:09 Instrument Air realigned to Containment.  
03:08 Commenced Reactor Coolant System (RCS) Cooldown  
04:20 Following troubleshooting and breaker replacement, closed Breaker LA08-02 and restored Letdown (Exit TS 3.8.9)  
04:34 NRC notified per 10 CFR 50.72 (EN# 56971)  
04:54 PZR level restored (Exit TS 3.4.9)

Following the reactor trip, main feedwater [EIS: JB] continued to be available. An actuation of the Auxiliary Feedwater System [EIS: BA] occurred following the manual reactor trip as designed due to low level in the steam generators [EIS: SB/SG].

**BASIS FOR REPORTABILITY AND SAFETY ASSESSMENT:**

The safety consequences of this event were low. Although the component failure had downstream interrelation system effects on letdown and makeup, the operating crew appropriately mitigated the transient with abnormal and emergency operating procedures. Additionally, all TS required actions were met and parameters restored well within required action times. The operating crew responded appropriately to the event. This event was within the analysis of the UFSAR Chapter 15. This event is reportable per 10 CFR 50.73(a)(2)(iv)(A) due to the automatic actuation of the Reactor Protection System [EIS: AA] and AFW system as listed in 10 CFR 50.73(a)(2)(iv)(B). FNP Unit 1 was not affected during this event.

**CAUSE:**

The 125 Volt DC AKW breaker was built by Siemens and equipped with SIGMATRIP DC Analog Trip Unit built by SURE-TRIP. Framatome was the vendor used to qualify the safety related breakers and trip units. The analysis determined that the vendor QA/QC program was not sufficient to identify poor practices when interfacing with third party products. It was determined that during installation of the SIGMATRIP DC Analog Trip Unit that the stranded wires that were terminated in the harness protruded in such a manner they were able to short to an adjacent wire in the harness. The short occurred in breaker LA08-2 (Serial Number R-300950258102B-001) on the A phase current transformer (CT)/sensor harness. For the SIGMATRIP DC Analog Trip Unit, Siemens and Framatome do not currently have a verification step/QA/QC check or any instruction for inspecting the wires connected in the wiring harnesses mounted to the CTs/sensors or the trip unit/logic box.

**LICENSEE EVENT REPORT (LER)  
CONTINUATION SHEET**

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**1. FACILITY NAME**

Joseph M. Farley Nuclear Plant, Unit 2

**050****052****2. DOCKET NUMBER**

364

**3. LER NUMBER****YEAR**

2024

**SEQUENTIAL  
NUMBER**

002

**REV  
NO.**

01

**NARRATIVE****CORRECTIVE ACTIONS:**

The associated breaker was immediately replaced by a spare breaker during the event. A systematic plan has been developed to inspect all AKW Series breakers which are installed or will be installed in the plant. Additionally, site actions have been created to monitor the changes by SURE-TRIP, Siemens, and Framatome associated with wire stripping, installation, inspection, and QC/QA checks.

Additionally, Framatome has notified SNC by letter dated March 26, 2024 (LTR24010) that their evaluation has determined that this results in a Part 21 Deviation because the potential failure to inadvertently trip Siemens Low Voltage (LV) AKW circuit breakers is a departure from the technical requirements included in the Procurement Document.

Framatome listed two Farley Purchase Orders with the same style AKW circuit breakers that were potentially impacted.

Based upon the best available information regarding the breaker failure at the time that U2 LER 2024-002-00 was initially submitted and preliminary results from the vendor's inspection of the other potentially impacted breakers listed in the transfer notice, Farley appropriately reported the issue per 10 CFR 50.73 as allowed by 10 CFR 21.2(c).

Farley has completed a Substantial Safety Hazard evaluation for internal documentation purposes and to provide reassurance that the impacted breakers would be reportable as a defect under 10 CFR 21, and that conclusion has been validated. Specifically, the evaluation determined that utilizing an impacted spare breaker in the most limiting application (Unit 1 B-Train Batteries or Unit 2 A-Train Batteries) could result in a complete failure of the DC buses and loss of safety function if the impacted breaker were to fail in conjunction with a single breaker failure on the redundant (opposite) train.

Basic Component: Siemens Low Voltage AKW Circuit Breaker

Serial Number: R-300950258102B-001

Supplying/Dedicating Entity: Framatome Inc.

**PREVIOUS SIMILAR EVENTS:**

No similar events or failure of AKW breakers were identified.